A Face-Projection Spin(10) Effective Framework: Rubik Covariance, $O(2)$ Family Geometry, and the Conditional GUT Boundary

Working draft

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Abstract

This draft rewrites the theory around a face-projection principle. Low-energy Standard-Model labels are not treated as primitive entries in a table; they are the projections of a higher-dimensional internal representation under a chosen symmetry-breaking face. In this language, one Spin(10) half-spinor 16 is the standard minimal six-face realization whose Standard-Model projection displays

$$Q, \quad L, \quad u^c, \quad d^c, \quad \nu^c, \quad e^c.$$

Three families are protected separately by the index

$$H^0(\mathbb{CP}^1, \mathcal{O}(2)), \qquad \dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3.$$

The result is not an unconditional PSLT-only derivation of a unique grand unified theory. It is a conditional Spin(10) effective framework. The Majorana contact tensor direction is no longer left arbitrary: it is the unique $SL(2)$-invariant second-transvectant direction K_{tr}. The paper also records a Route-B candidate: a visible-singlet sterile transvectant messenger can generate the ζK_{tr} contact by a Schur complement when $\lambda^2 = \zeta$. The remaining open datum is the hidden phase/radial sector that supplies this complex coefficient, together with companion audits for flavor, proton decay, thresholds, and source-sector UV origin. Detailed machine-audit and scan appendices have been removed from this version until it is clear which of them truly belong to the face-projection framework.

1 Scope of the Lean Route-A Note

This version is deliberately a Route-A paper skeleton. Its theorem-level claims are limited to three representation-geometric statements:

six faces	Spin(10) half-spinor restriction gives one SM family plus ν^c,
three copies	$\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ Dolbeault index gives three protected sections,
contact direction	K_{tr} is the unique $SL(2)$ second-transvectant direction.

Route B is kept only as a conditional candidate mechanism for the complex coefficient ζ. Full CKM/PMNS fitting, dimension-five proton-decay Wilson tensors, threshold matching, and source-sector UV completion are not claimed in this lean note; they remain companion-audit problems. Thus the paper should be read as a conditional face-projection Spin(10) EFT note, not as a completed first-principles derivation of a unique GUT.

2 Face Projection

The organizing idea is that the visible particle table is a projection table, not a table of fundamental labels. Let G be a high-energy compact group, let R be a complex representation of G,

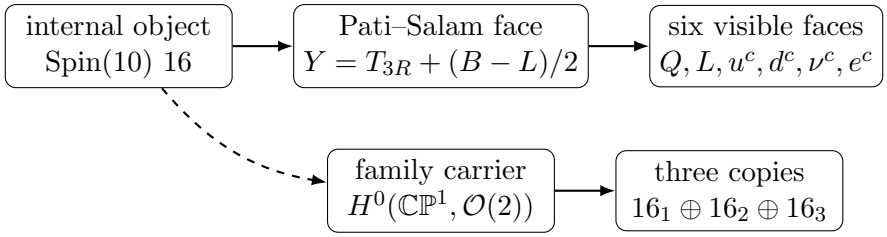

Figure 1: Face-projection picture. The six Standard-Model multiplet types are visible faces of one Spin(10) half-spinor, while the family number is an independent protected index.

and let a vacuum orientation θ select a broken subgroup

$$H_\theta \simeq \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y.$$

The face projection is the restriction functor

$$\Pi_\theta(R) := \mathrm{Res}^G_{H_\theta} R = \bigoplus_\alpha m_{\alpha,\theta}\, r_{\alpha,\theta},$$

where the irreducible H_θ-summands $r_{\alpha,\theta}$ are the visible multiplet types. If $g \in G$, then the embedded subgroup changes by conjugation,

$$H_\theta \longmapsto H_{g\theta} = g H_\theta g^{-1}.$$

The same high-energy representation can therefore display different visible labels under a different face:

$$\Pi_{g\theta}(R) = \mathrm{Res}^G_{gH_\theta g^{-1}} R.$$

This is the representation-theoretic version of the intuitive statement that the same internal object can look different after a legal internal rotation.

Definition 2.1 (Rubik covariance)**.** A unified model is face-covariant if the visible low-energy multiplet types are irreducible summands in restrictions of a common high-energy representation, and if every allowed change of face is a conjugation of the broken-subgroup embedding preserving anomaly cancellation, index, chirality, charge normalization, and representation dimension. The principle does not allow arbitrary transmutation of low-energy particles. It says that different low-energy labels can be different visible faces of the same high-energy object.

The Rubik-cube image is useful only in this precise sense. A chemical element is still defined by nuclear charge; it does not become another element because one rotates a drawing. The analogy points instead to root lattices, Weyl chambers, gauge-basis rotations, and symmetry-breaking projections.

The concrete grand-unification branch used below follows the standard Pati–Salam and Spin(10) route [1, 2, 3, 4]. The conditional neutrino sector uses the usual type-I seesaw logic [5, 6, 7], while the threshold language is the conventional perturbative RGE language [8, 9].

3 The Six-Face Spin(10) Object

Proposition 3.1 (Standard minimal six-face realization)**.** *Among standard simple compact GUT candidates, if the six chiral Standard-Model multiplet types*

$$Q, \quad L, \quad u^c, \quad d^c, \quad \nu^c, \quad e^c$$

are required to arise from one irreducible complex chiral representation whose restriction contains exactly one SM family plus ν^c, with no chiral SM exotics and with minimal representation dimension, the standard minimal single-object realization is the Spin(10) *half-spinor* 16*.*

Proof. The Pati–Salam subgroup of Spin(10) gives

$$16 \to (4,2,1) \oplus (\bar{4},1,2).$$

Breaking

$$\mathrm{SU}(4)_C \to \mathrm{SU}(3)_C \times \mathrm{U}(1)_{B-L}$$

gives

$$4 \to (3,1/3) \oplus (1,-1), \qquad \bar{4} \to (\bar{3},-1/3) \oplus (1,+1).$$

With

$$Y = T_{3R} + \frac{B-L}{2},$$

the restriction to the Standard-Model face is

$$\mathrm{Res}^{\mathrm{Spin}(10)}_{G_{\mathrm{SM}}}\, 16 = Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c,$$

where the six projected Standard-Model multiplet types are

field	$\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$
Q	$(3,2)_{1/6}$
L	$(1,2)_{-1/2}$
u^c	$(\bar{3},1)_{-2/3}$
d^c	$(\bar{3},1)_{1/3}$
ν^c	$(1,1)_0$
e^c	$(1,1)_1$.

The $SU(5)$ alternative uses $10 \oplus \bar{5} \oplus 1$, which is not one irreducible chiral object. Pati–Salam is natural but not simple. E_6 contains this structure inside a larger 27, usually bringing extra matter that must be lifted. Therefore Spin(10) with a 16 is the standard minimal single-object realization. □

Proposition 3.2 (Hypercharge normalization)**.** *The face projection*

$$Y = T_{3R} + \frac{B-L}{2}$$

has the GUT normalization

$$\frac{\mathrm{Tr}\, Y^2}{\mathrm{Tr}\, T_{3L}^2} = \frac{5}{3}.$$

Proof. The trace is over one left-handed 16 after Standard-Model decomposition. For one family,

$$\mathrm{Tr}\, Y^2 = 6\left(\frac{1}{6}\right)^2 + 2\left(-\frac{1}{2}\right)^2 + 3\left(-\frac{2}{3}\right)^2 + 3\left(\frac{1}{3}\right)^2 + 1(1)^2 + 1(0)^2 = \frac{10}{3}.$$

The $SU(2)_L$ doublets are Q and L, and the doublet normalization is $\mathrm{Tr}_{\mathbf{2}}\, T_3^2 = 1/2$. Hence

$$\mathrm{Tr}\, T_{3L}^2 = 3 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = 2.$$

Thus

$$\frac{\mathrm{Tr}\, Y^2}{\mathrm{Tr}\, T_{3L}^2} = \frac{10/3}{2} = \frac{5}{3}.$$

□

Remark 3.3 (What rotation can and cannot mean). At low energy Q cannot be freely rotated into e^c. The Standard-Model gauge symmetry has already selected superselection-like labels. The face-projection statement is about the high-energy representation and its broken-subgroup projection. Any physical process that changes the visible face must be mediated by legitimate high-energy fields and is constrained, for example, by proton decay.

4 Three Families from an $O(2)$ Index

The family number is not produced by rotating the six-face object. It is a separate index statement. The analytic input is the standard Spin^c Dolbeault–Dirac/cohomology identification and the Riemann–Roch–Serre-duality calculation on $\mathbb{CP}^1$ [10, 11].

Theorem 4.1 (Conditional Spin^c index theorem for three protected families)**.** *Assume that the protected internal family carrier is the* Spin^c *Dolbeault bundle on*

$$C = \mathbb{CP}^1$$

twisted by a two-center divisor

$$R = p_+ + p_-, \qquad L_R = \mathcal{O}(R) \simeq \mathcal{O}(2).$$

Assume also that all other SM-charged internal sectors are absent, gapped, or vectorlike, so they contribute no additional unpaired chiral zero modes. Let

$$D_R = \sqrt{2}\left(\bar{\partial}_{L_R} + \bar{\partial}^\dagger_{L_R}\right)$$

be the corresponding Spin^c *Dolbeault–Dirac operator. Then*

$$\ker D_R^+ \simeq H^0(\mathbb{CP}^1, \mathcal{O}(2)), \qquad \ker D_R^- \simeq H^1(\mathbb{CP}^1, \mathcal{O}(2)),$$

with

$$\dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3,$$

and $H^1(\mathbb{CP}^1, \mathcal{O}(2)) = 0$*. Thus the protected sector contributes exactly three positive-chirality family zero modes.*

Proof. Hodge theory identifies the positive and negative Spin^c Dolbeault–Dirac kernels with the Dolbeault cohomology groups displayed above. Riemann–Roch on $\mathbb{CP}^1$ gives

$$h^0(\mathbb{CP}^1, \mathcal{O}(2)) - h^1(\mathbb{CP}^1, \mathcal{O}(2)) = 2 + 1 = 3.$$

Serre duality gives

$$h^1(\mathbb{CP}^1, \mathcal{O}(2)) = h^0(\mathbb{CP}^1, K_{\mathbb{CP}^1} \otimes \mathcal{O}(-2)) = h^0(\mathbb{CP}^1, \mathcal{O}(-4)) = 0.$$

Therefore $h^0 = 3$. □

Corollary 4.2 (Projected chiral state space)**.** *Under the six-face* $\mathrm{Spin}(10)$ *assumption and the* $O(2)$ *family carrier, the low-energy chiral state space is*

$$\mathcal{S}^{4D}_{\mathrm{chiral}} = 16 \otimes H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq 16_1 \oplus 16_2 \oplus 16_3.$$

Remark 4.3 (Why this is still conditional). This theorem does not derive $C = \mathbb{CP}^1$, the two-center divisor R, the Spin^c carrier, or the no-extra-unpaired-sector condition. Those are branch assumptions. The theorem says that once those inputs are chosen, the family count is protected and is not a tunable scan output.

5 Conditional EFT Data

The face-projection and index principles supply the representational skeleton. They do not by themselves determine the full action. The current conditional branch retains the following action-level data:

A1 Spin(10) face-projection EFT with complete 16_i,
A2 $H^0(\mathbb{CP}^1, \mathcal{O}(2))$ protects three copies of the 16,
A3 type-I seesaw with a source-fixed Majorana contact,
A4 proton bounds and two-loop threshold matching,
A5 constrained/composite 54/210 source order parameters,
A6 auxiliary/composite conormal multipliers,
A7 a Schur lock with singlet auxiliaries and sterile completion.

Lemma 5.1 (Unique $SL(2)$-invariant contact direction)**.** *Let*

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \mathrm{Sym}^2\mathbb{C}^2.$$

Then

$$\mathrm{Sym}^2 V \simeq \mathrm{Sym}^4\mathbb{C}^2 \oplus \mathrm{Sym}^0\mathbb{C}^2.$$

The Veronese moment sector is the $\mathrm{Sym}^4\mathbb{C}^2$ *summand, while the orthogonal singlet contact sector is unique up to phase. In the normalized* (ψ_0, ψ_1, ψ_2) *basis used for the family carrier, this singlet can be represented as*

$$K_{\mathrm{tr}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Coordinate-free, $K_{\mathrm{tr}} \in \mathrm{Sym}^2 V^\vee$ *and its inverse bilinear form is* $K_{\mathrm{tr}}^{-1} \in \mathrm{Sym}^2 V$*. In the displayed orthonormal basis, the induced matrix obeys*

$$K_{\mathrm{tr}}^2 = \frac{1}{3} I, \qquad K_{\mathrm{tr}}^{-1} = 3K_{\mathrm{tr}}.$$

Proof. This is the Clebsch–Gordan decomposition for $\mathrm{Sym}^2(\mathrm{Sym}^2\mathbb{C}^2)$ [12]. The scalar summand has multiplicity one, so any invariant symmetric contact tensor is proportional to the displayed second-transvectant matrix. □

Benchmark Datum 5.2 (Majorana contact decomposition of the current benchmark)**.** The source-consistent Majorana matrix in the current benchmark is not a pure $(\mathbb{CP}^1, \mathcal{O}(2))$ Veronese matrix. It requires the unique orthogonal second-transvectant contact component

$$\frac{M_R}{M_*} = M_V + \zeta K_{\mathrm{tr}},$$

with

$$\zeta = 0.1076472949 + 0.0736514853\, i.$$

The contact fraction is

$$\frac{\|\zeta K_{\mathrm{tr}}\|}{\|M_R/M_*\|} = 1.304275 \times 10^{-1}.$$

The companion machine audit that produced the previous long draft decomposes the inverse-seesaw reconstructed M_R into the Veronese subspace plus the orthogonal trace/contact direction. The Veronese-plus-contact residual is at machine precision, whereas the Veronese-only residual equals the displayed contact fraction. This means the contact is not a negligible correction to the present PMNS-compatible branch. The lean note records the datum and the geometrically fixed tensor direction; the full matrices, basis cards, and norm conventions are deferred to companion artifacts.

Observation 5.3 (Negative scan for simple hidden-sector origins of ζ)**.** Within the currently tested simple hidden-quotient and small monomial clockwork classes, ζ is a conditional datum rather than a derived number.

The scan evidence is limited but sharp within that tested class. The contact-sensitivity scan gives the loose PMNS-compatible windows

$$\Delta s = 3.208798 \times 10^{-5}, \qquad \Delta\delta = 5.424119 \times 10^{-5} \text{ rad}.$$

Removing the contact spoils the PMNS and mass-splitting replay. A minimal hidden D-quotient leaves modulus and phase flat directions. A small real-coefficient monomial or clockwork phase with denominator $N \leq 6$ misses the target phase by at least

$$4.471595 \times 10^{-1} \text{ rad},$$

and the first root of unity inside the loose phase window appears only at

$$N = 178.$$

Thus no small hidden quotient or small monomial clockwork mechanism in the current evidence set predicts the complex value of ζ. A future non-minimal hidden sector may still do so, but it would be a new assumption that must be audited with its threshold and UV cost.

6 Route B: Hidden Transvectant Origin for ζ

The negative scan above does not forbid a hidden origin for ζ. It only says that the currently tested small hidden quotients and small monomial clockwork classes do not predict it. The conservative Route-B move is therefore to add a new hidden-sector theorem: derive the tensor direction from the unique transvectant K_{tr}, and isolate the remaining burden in a hidden phase/radial sector that supplies the complex number.

Theorem 6.1 (Post-$B-L$ sterile transvectant messenger)**.** *Let*

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)).$$

After $B-L$ breaking, let

$$N_i = P_{\nu^c}(16_i), \qquad N \in V,$$

be the right-handed neutrino family vector selected by the Majorana source projector. Let

$$X \in V^\vee$$

be a visible-singlet sterile family messenger. At the post-$B-L$ matching scale, consider the coordinate-free superpotential

$$\frac{W_R}{M_*} = \frac{1}{2} M_V(N,N) + \lambda X(N) - \frac{1}{2} K_{\text{tr}}^{-1}(X,X),$$

where

$$M_V \in \text{Sym}^2 V^\vee, \qquad K_{\text{tr}} \in \text{Sym}^2 V^\vee, \qquad K_{\text{tr}}^{-1} \in \text{Sym}^2 V.$$

In a chosen basis this is

$$\frac{W_R}{M_*} = \frac{1}{2} N^T M_V N + \lambda X^T N - \frac{1}{2} X^T K_{\text{tr}}^{-1} X,$$

where K_{tr} is the unique $SL(2)$-invariant contact direction from Lemma 4.1. Integrating out X gives

$$\frac{W_R^{\text{eff}}}{M_*} = \frac{1}{2} \left(M_V + \lambda^2 K_{\text{tr}}\right)(N,N).$$

Thus the benchmark contact is generated if

$$\lambda^2 = \zeta.$$

Proof. The X-F-term is the equality in V

$$\frac{\partial W_R}{\partial X} = M_* \left(\lambda N - K_{\rm tr}^{-1} X\right) = 0.$$

Here $K_{\rm tr}^{-1} : V^\vee \to V$ is the inverse map induced by the non-degenerate bilinear form $K_{\rm tr}$. Thus

$$X = \lambda K_{\rm tr} N,$$

where $K_{\rm tr} : V \to V^\vee$. Substitution gives, in any basis,

$$\begin{aligned}
\frac{W_R^{\rm eff}}{M_*} &= \frac{1}{2} M_V(N,N) + \lambda X(N) - \frac{1}{2} K_{\rm tr}^{-1}(X,X) \\
&= \frac{1}{2} M_V(N,N) + \lambda^2 K_{\rm tr}(N,N) - \frac{1}{2}\lambda^2 K_{\rm tr}(N,N) \\
&= \frac{1}{2}\left(M_V + \lambda^2 K_{\rm tr}\right)(N,N).
\end{aligned}$$

Equivalently, in matrix notation,

$$\frac{W_R^{\rm eff}}{M_*} = \frac{1}{2} N^T M_V N + \frac{1}{2}\lambda^2 N^T K_{\rm tr} N.$$

This is exactly the displayed effective Majorana matrix. □

Remark 6.2 (Scope of the projector). The field $N_i = P_{\nu^c}(16_i)$ is a post-$B-L$-breaking EFT variable. Before symmetry breaking, a Spin(10)-singlet messenger cannot select ν^c from the 16_i without an additional Majorana source or projector. Route B therefore assumes such a post-breaking projector, for example from a $(B-L) = -2$ Majorana source sector, rather than deriving it from face projection alone. In this lean note P_{ν^c} is therefore an input of the post-$B-L$ EFT matching problem, not a result of the face-projection principle.

Benchmark Datum 6.3 (Target coefficient and square-root coupling)**.** For the current benchmark,

$$\zeta_{\rm tar} = 0.1076472949 + 0.0736514853\, i.$$

Its polar data are

$$|\zeta_{\rm tar}| = 0.13043190325293763, \qquad \arg \zeta_{\rm tar} = 0.600038020318215.$$

One square-root coupling is

$$\lambda = \sqrt{\zeta_{\rm tar}} = 0.34502115743308964 + 0.10673473744038917\, i,$$

with

$$|\lambda| = 0.3611535729477664.$$

This is a moderate hidden coupling; it is not a high-power clockwork suppression.

Observation 6.4 ($\mathbb{Z}_{178}$ phase diagnostic)**.** A discrete phase can approximate the benchmark phase by

$$\zeta_{178} = |\zeta_{\rm tar}| \exp\left(\frac{2\pi i\, 17}{178}\right).$$

Since

$$\frac{2\pi\, 17}{178} = 0.6000794956295111,$$

the phase error is

$$4.1475311296 \times 10^{-5} \text{ rad},$$

which lies inside the loose phase window

$$5.424119 \times 10^{-5} \text{ rad.}$$

The direct complex error is

$$|\zeta_{178} - \zeta_{\text{tar}}| = 5.4097037899 \times 10^{-6}.$$

This does not make ζ an exact prediction. It supplies a precise discrete-phase candidate; exact equality would require either a small axion shift or a continuous hidden phase. The Route-B hidden-messenger theorem uses only $\lambda^2 = \zeta$; the $\mathbb{Z}_{178}$ value is kept as a side diagnostic, not as a premise of the benchmark branch.

Observation 6.5 (Nonperturbative hidden-spurion option)**.** An alternative is

$$\zeta = Ae^{-S_H + i\theta_H}.$$

For $A = 1$,

$$S_H = -\log|\zeta_{\text{tar}}| = 2.0369040025655094, \qquad \theta_H = \arg \zeta_{\text{tar}}.$$

This is naturally interpreted as a nonperturbative hidden-sector spurion. It is a conditional physical origin, not a weak-coupling derivation from face-projection alone.

Lemma 6.6 (Minimal post-$B - L$ R-selection rule)**.** *At the post-$B - L$ matching scale, take the superpotential to have $R(W) = 2$. A minimal selection rule that keeps the sterile messenger Majorana-only is*

object	*role*	$U(1)_R$
16_i *components*	*matter multiplets, including* N_i	1
$H, T, \bar{T}$	*Higgs doublet/triplet fields*	0
$X \in V^\vee$	*sterile transvectant messenger*	1
$\lambda,\ M_V,\ K_{\text{tr}}^{-1}$	*matching tensors/spurions*	0

together with the post-breaking projector

$$N_i = P_{\nu^c}(16_i).$$

Then the Route-B Majorana terms

$$X(N), \qquad K_{\text{tr}}^{-1}(X, X), \qquad M_V(N, N)$$

all have $R = 2$, while ordinary Dirac and triplet Yukawa terms

$$16_i 16_j H, \qquad 16_i 16_j T$$

also have $R = 2$. However any superpotential operator with at least one additional X inserted into a Dirac or colored-triplet operator has

$$R = 2 + m, \qquad m \geq 1,$$

and is therefore forbidden if the R-selection rule is exact at the matching scale.

Proof. The allowed Route-B terms have

$$R(X) + R(N) = 1 + 1 = 2, \qquad 2R(X) = 2, \qquad 2R(N) = 2.$$

The baseline Dirac and triplet Yukawa terms have

$$R(16_i) + R(16_j) + R(H \text{ or } T) = 1 + 1 + 0 = 2.$$

Adding $m \geq 1$ sterile messengers to such an operator shifts the superpotential charge to $2 + m$, so the operator is not allowed. This proves the selection rule at the holomorphic matching level. If later R-breaking spurions are introduced, they must be charged so that these forbidden operators remain absent to the order being audited. □

Remark 6.7 (Status of the R-selection rule). The $U(1)_R$ assignment in Lemma 6.6 is used here as a holomorphic EFT matching selection rule. It is not yet claimed to be an anomaly-free microscopic symmetry. A UV implementation would require either an anomaly-free continuous or discrete R-symmetry, or a spurionic origin whose R-breaking insertions do not regenerate X-decorated Dirac or colored-triplet operators within the audited order.

Theorem 6.8 (Tree-level canonical matching silence of Majorana-only hidden contact)**.** *In a canonical-normalized post-$B-L$ EFT at the matching scale, use Route B as a post-$B-L$-breaking matching theorem, with*

$$N_i = P_{\nu^c}(16_i)$$

as above. Assume the hidden messenger sector obeys the R-selection rule of Lemma 6.6:

$$X \text{ is a visible singlet}, \qquad X \text{ couples only to the Majorana channel } K_{\mathrm{tr}}(N,N),$$

and X has no couplings to Dirac Yukawa operators or colored-triplet operators. If integrating out X gives

$$\Delta M_R/M_* = \zeta K_{\mathrm{tr}},$$

then the visible one-loop threshold vector, the Dirac Yukawa matrices, and the dimension-five colored-triplet Wilson tensors are unchanged relative to the benchmark.

Proof. Since X is a visible singlet,

$$\Delta b_1^X = \Delta b_2^X = \Delta b_3^X = 0,$$

so it contributes no non-universal visible one-loop threshold. The Majorana-only selection rule gives

$$\delta Y_u = \delta Y_d = \delta Y_e = \delta Y_{\nu D} = 0.$$

The same rule forbids hidden insertions in the colored-triplet Yukawa couplings and inverse triplet propagator, so

$$\delta C_{\bar{5}L} = \delta C_{\bar{5}R} = 0.$$

If ζ equals the benchmark value, then M_R is the benchmark Majorana matrix and the type-I seesaw replay is unchanged. Thus the hidden contact can generate the missing Majorana tensor without moving the visible flavor, proton-decay, or one-loop threshold audits. □

Observation 6.9 (Canonical basis caveat)**.** The previous theorem is a tree-level matching statement in a canonical-normalized EFT basis. A microscopic completion with a finite messenger interval can generate a right-handed-neutrino wave-function correction of order

$$\delta Z_N \sim \frac{|\lambda|^2}{16\pi^2} \log \frac{M_*}{M_X}.$$

For the benchmark value,

$$\frac{|\lambda|^2}{16\pi^2} = 8.259697 \times 10^{-4}.$$

This is small, but not automatically smaller than the narrow contact-sensitivity windows. Therefore the benchmark matrices in this lean note are defined after canonical normalization. If a microscopic model predicts a non-negligible δZ_N, the companion seesaw replay must use

$$Y_{\nu D} \mapsto Y_{\nu D}\left(1 - \frac{1}{2}\delta Z_N\right),$$

$$M_R \mapsto M_R - \frac{1}{2}\left(\delta Z_N^T M_R + M_R \delta Z_N\right),$$

before comparing to the PMNS benchmark.

check	value	role
$\lvert\zeta_{\rm tar}\rvert$	0.13043190325293763	target modulus
$\arg\zeta_{\rm tar}$	0.600038020318215	target phase
$\sqrt{\zeta_{\rm tar}}$	$0.345021 + 0.106735\,i$	hidden messenger coupling
$\lvert\lambda\rvert$	0.3611535729477664	moderate coupling
$\lVert\lambda^2 K_{\rm tr} - \zeta K_{\rm tr}\rVert_F$	1.20185×10^{-17}	Schur-complement match
$\lVert K_{\rm tr}(3K_{\rm tr}) - I\rVert_F$	3.84593×10^{-16}	inverse convention
$2\pi(17/178) - \arg\zeta$	4.14753×10^{-5} rad	side phase diagnostic
$\lvert\zeta_{178} - \zeta_{\rm tar}\rvert$	5.40970×10^{-6}	discrete approximation
$-\log\lvert\zeta\rvert$	2.0369040025655094	instanton-spurion option
$\lvert\lambda\rvert^2/(16\pi^2)$	8.25970×10^{-4}	canonical-basis audit scale
$\Delta b^X_{\rm vis}$	$(0, 0, 0)$	visible threshold silence

Table 1: Compact Route-B numerical verification. The table records only the algebraic checks used in the lean hidden-ζ section; it is not a replacement for the deferred full flavor/proton/threshold companion audits.

7 Theorem Boundary

Theorem 7.1 (Conditional benchmark boundary)**.** *Given the face-projection data, the $O(2)$ index data, the companion benchmark audits, and either a benchmark-specified Majorana datum $\mathcal{D}_\zeta$ or a Route-B hidden messenger producing $\mathcal{D}_{\text{hidden zeta}}$, the current construction defines one internally checked conditional EFT branch:*

$$\frac{\mathcal{D}_{\rm face} + \mathcal{D}_{\rm index} + (\mathcal{D}_\zeta \text{ or } \mathcal{D}_{\text{hidden zeta}}) + \mathcal{D}_{\text{companion audits}}}{\rightsquigarrow \text{ one conditional benchmark branch.}}$$

This does not support the stronger implication

$$\textit{PSLT/Another Physics alone} \Longrightarrow \textit{a unique GUT with predicted } \zeta.$$

Proof. A complete microscopic GUT requires an explicit gauge group, chiral matter representations, anomaly cancellation, a breaking potential, Yukawa and Majorana sectors, baryon-violating operators, and threshold matching. Face projection motivates the Spin(10) representation geometry. The $O(2)$ index protects the family count. Neither supplies all action-level data. The remaining branch data A1–A7 and the benchmark-level checks are conditional inputs or companion artifacts, and they are required whether ζ is supplied as a benchmark datum or generated through Route B. The Route-B messenger derives $\zeta K_{\rm tr}$ only after adding a hidden sterile messenger, phase/radial data, and Majorana-only selection rules. It therefore improves the conditional EFT explanation of ζ, but it is still not a PSLT-only prediction and it does not replace the companion phenomenological audits. □

Corollary 7.2 (Status of the rewritten theory)**.** *The complete claim of this lean version is*

$$\boxed{\begin{array}{c}\textit{face-projection motivation} + \textit{ conditional } \mathrm{Spin}(10) \textit{ EFT} \\ \neq \textit{unconditional PSLT-only GUT proof.}\end{array}}$$

The rewrite is nevertheless useful: it gives a coherent representation-level reason why Spin(10) *is the natural minimal six-face family object and why three families are protected by an independent $O(2)$ index.*

8 Lean-Note Status Ledger

The logical status of this version is deliberately narrower than a complete GUT construction:

status	**content**
proved/constructed	Spin(10) face skeleton; $\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ index; K_{tr} uniqueness; Route-B Schur complement.
conditional inputs	$C = \mathbb{CP}^1$, $R = p_+ + p_-$; protected Spin^c carrier; no unpaired SM-charged exotics; P_{ν^c}; R-selection rule; hidden phase/radial sector.
deferred audits	full flavor fit; $d = 5$ proton-decay Wilson tensors; threshold/proton companion audits; source-sector UV origin.
not claimed	PSLT-only unique GUT proof.

9 Minimal Reproducibility Manifest

The lean note has a small, explicit reproducibility surface. It does not include the deferred full flavor/proton/threshold scans.

item	**artifact or command**
paper source	`paper/gut_framework.tex`, `paper/refs.bib`
compiled note	`paper/gut_framework.pdf`
D5 appendix check	`python3 code/verify_d5_half_spin_hypercube.py`
D5 outputs	D5 summary, weights, and report in `output/d5_half_spin/`
Route-B algebra check	`python3 code/verify_hidden_zeta_origin.py`
Route-B output	verification JSON in `output/hidden_zeta_origin/`
status log	`roadmap.md`

The reproducible numerical claims inside this TeX draft are therefore only the D5 half-spin branching check, the hypercharge normalization check, and the Route-B algebraic ζK_{tr} matching check. All publication-grade phenomenology remains outside this manifest until the companion audits are restored.

10 Deferred Calculations

The previous version contained detailed scan and audit material for Yukawa textures, $d = 5$ proton decay, threshold corrections, source sectors, clockwork rescue, no-web provenance, and theorem ledgers. Those calculations are not deleted from the project history: they are preserved in the backup made before this rewrite and in the corresponding output artifacts. They are omitted from this TeX draft because their status relative to the face-projection principle is not yet settled. This lean note defines the representation-level and index-level skeleton; benchmark-level numerical checks are deferred to companion artifacts. The only numerical Route-B checks kept in this draft are the algebraic ζ, K_{tr}, and phase-spurion checks needed to support the hidden messenger theorem.

The conservative paper strategy is Route A as the main note: keep the proved representation and index skeleton in the foreground, and treat Route B as a candidate hidden-sector mechanism or companion-paper seed. Promoting Route B to the main line would require restoring only those flavor, proton-decay, threshold, and source-sector calculations that survive the R-selection rule and canonical-normalization replay. In short,

Route A	main paper: conditional face-projection model note,
Route B	candidate mechanism or companion: hidden origin for ζ.

11 Conclusion

The clean result of the rewrite is a bounded theorem stack. A Spin(10) half-spinor supplies a standard minimal six-face object; the $\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ carrier supplies three protected copies; and the second-transvectant singlet supplies the unique Majorana contact direction. These are mathematical inputs and consequences inside a conditional EFT branch. They do not by themselves determine the microscopic source sector, the complex hidden coefficient ζ, the full flavor fit, or the proton-decay Wilson tensors. Keeping this boundary explicit is the point of the lean Route-A version.

A D_5 Half-Spin Weight Hypercube

This appendix records the compact representation-theoretic calculation behind the six-face statement. It is not a new uniqueness theorem; it is the weight-level explanation of the standard Spin(10) half-spinor branch used in the main text [4, 12].

Let $e_1, \ldots, e_5$ be an orthonormal basis for the D_5 weight space. The roots of D_5 are

$$\pm e_a \pm e_b, \qquad a \neq b.$$

Choose the positive-chirality half-spin representation to have weights

$$\Lambda_{16} = \left\{ \frac{1}{2} \sum_{a=1}^{5} s_a e_a : s_a = \pm 1, \quad \#\{a : s_a = -1\} \equiv 0 \pmod 2 \right\}.$$

There are $2^{5-1} = 16$ such weights. The Pati–Salam face is the split

$$D_5 \supset D_3 \oplus D_2, \qquad \mathrm{Spin}(6) \times \mathrm{Spin}(4) \simeq \mathrm{SU}(4)_C \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R.$$

Use s_1, s_2, s_3 for the D_3 spinor weights and s_4, s_5 for the D_2 spinor weights. Define

$$m_C = \#\{a \leq 3 : s_a = -1\}, \qquad m_W = \#\{a = 4, 5 : s_a = -1\}.$$

The chirality condition says $m_C + m_W$ is even, so m_C and m_W have the same parity. Therefore the weights split into exactly two blocks:

m_C parity	m_W parity	number of weights	Pati–Salam block
even	even	$4 \cdot 2 = 8$	$(4, 2, 1)$
odd	odd	$4 \cdot 2 = 8$	$(\bar{4}, 1, 2)$.

Thus

$$16 \to (4, 2, 1) \oplus (\bar{4}, 1, 2).$$

One convenient convention for the $\mathrm{SU}(4)_C$ component is

first three signs	$\mathrm{SU}(3)_C$	$B - L$
zero minus signs	1	-1
two minus signs	3	$1/3$
one minus sign	$\bar{3}$	$-1/3$
three minus signs	1	1.

For the weak D_2 signs, take

$$(s_4, s_5) = (+, +), (-, -) \quad \Longrightarrow \quad \mathrm{SU}(2)_L \text{ doublet}, \quad T_{3L} = \frac{s_4 + s_5}{4}, \quad T_{3R} = 0,$$

and

$$(s_4, s_5) = (+,-), (-,+) \quad \Longrightarrow \quad \mathrm{SU}(2)_R \text{ doublet}, \quad T_{3R} = \frac{s_4 - s_5}{4}, \quad T_{3L} = 0.$$

Applying

$$Y = T_{3R} + \frac{B-L}{2}$$

then gives the six visible Standard-Model faces:

field	$\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L$	Y	multiplicity
Q	$(3,2)$	$1/6$	6
L	$(1,2)$	$-1/2$	2
u^c	$(\bar{3},1)$	$-2/3$	3
d^c	$(\bar{3},1)$	$1/3$	3
ν^c	$(1,1)$	0	1
e^c	$(1,1)$	1	1.

The multiplicities sum to 16, so this is a complete half-spinor, not a collection of separately chosen low-energy fragments.

The appendix calculation is reproduced by

`code/verify_d5_half_spin_hypercube.py`.

The generated summary reports

$$\#\Lambda_{16} = 16, \qquad (4,2,1) : 8, \qquad (\bar{4},1,2) : 8,$$

the six field multiplicities

$$(Q, L, u^c, d^c, \nu^c, e^c) = (6,2,3,3,1,1),$$

and the same normalization check as in the main text,

$$\mathrm{Tr}\, Y^2 = \frac{10}{3}, \qquad \mathrm{Tr}\, T_{3L}^2 = 2, \qquad k_Y = \frac{5}{3}.$$

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